Neil Janwani

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TECHNICAL INTERESTS

Broadly, my interests are in **learning, control theory, legged robots, and human robot interaction (HRI)**. I am currently working on the following threads for my PhD thesis:

1. Learning robust and tuneable bipedal locomotion and behavior; fusing imitation learning with model-based control
2. Constructing multi-objective algorithms for preference learning, solving for pareto-optimal policy families
3. Some minor threads, including socially expressive humanoid controllers and VLM-based task planners

EDUCATION

Georgia Institute of Technology, Atlanta, Georgia, USA — AUG 2024 — JUN 2029
Ph.D. in Robotics

- Research Group: Dynamic Mobility Lab
- Research Advisor: Prof. Maegan L. Tucker
- Selected Courses: Interactive Robot Learning, Nonlinear Control, Human-Robot Interaction

California Institute of Technology, Pasadena, California, USA — SEP 2020 — JUN 2024
B.S. in Computer Science

- Research Advisors: Dr. Joel W. Burdick, Dr. Aaron D. Ames
- Selected Courses: Large Language and Vision Models, Nonlinear Dynamics and Control, Kinematics, Planning, Mobile Robots, Manipulators

RESEARCH

Preprints:

- **Janwani, N. C.***, Madabushi, V*., Tucker, M. (2025). "NaviGait: Navigating Dynamically Feasible Gait Libraries using Deep Reinforcement Learning."[Paper]

Publications:

- **Janwani, N. C.***, Daş, E*., Touma, T., Wei, S. X., Molnar, T. G., & Burdick, J. W. (2023). "A Learning-Based Framework for Safe Human-Robot Collaboration with Multiple Backup Control Barrier Functions." *IEEE International Conference on Robotics and Automation (ICRA), 2024* [Paper] [Video]

Patents:

- Shaikewitz L., Tucker M., **Janwani N.**, Ames A., "Front-leg assistive exoskeleton" *US20240245569A1* [Paper]

PROFESSIONAL EXPERIENCE

Dynamic Mobility, Georiga Tech — Atlanta, Georgia
PhD Student: IRIM and NSF Graduate Student Fellow — AUG 2024 — Present

- Developed NaviGait, a hierachical architecture written in JaX that fuses reinforcement learning with gait libraries to produce robust and tuneable policies capable of dynamic locomotion. Submitted to ICRA2026 in September.
- Designing PrefMORL, a multi-objective reinforcement learning framework that learns pareto-optimal policy families with respect to learned objectives. Written fullyin JaX
- Mujoco, Gazebo, etc with HLIP??

Burdick Group, Caltech — Pasadena, California
Research Fellow: Kiyo and Eiko Tomiyasu Named Scholar — JUN 2023 — Present

- Applied backup control barrier functions (BCBFs) to driver assistance by leveraging driver intention to choose between multiple BCBFs with different backup controllers.
- Fused LSTM architecture for intention estimation with BCBFs without losing theoretical safety guarantees.
- Implemented algorithms and collected data on hardware, and experimentally showed that human-robot alignment and general reachability of the robot was enhanced with my proposed method.
- Developed graphical and haptic user interface for 20+ DARPA team in accordance with team meetings and deadlines.

AMBER Lab, Caltech — Pasadena, California
Research Fellow — JAN 2021 — Present

- Developed a novel ankle exoskeleton built from a shin-mounted hand-shearing auxetic elastic actuator.
- Devised accurate LSTM and MLP-based gait phase approximators from data collected from ankle-mounted IMUs.
- Designed PCBs to house electronic components, optimally placed biometric sensors, and prioritized user safety.
- Programmed field oriented control and controller communication protocol for Maxon motors using SimpleFOC.

MIT: Lincoln Laboratory Lexington, Massachusetts
Research Intern JUN 2022 — SEP 2022

- Formulated additions to linear time-invariant control methodology for sensorless control of gimbal motor.
- Refactored Simulink and Jupyter API for experimentation of sensorless control algorithms on gimbal testbed.
- Achieved accurate sensorless estimation of steady-state electrical phase in under 20 seconds.

TEACHING EXPERIENCE

Single Board Computers in Research, Caltech CS 12 Course Pasadena, California
Course founder and instructor JAN 2023 — Present

- Developed 10-week course on prototyping robotic and computing systems for research applications.
- Taught 50 students in concurrent software and circuit design, sensor interfacing, microcontroller communication.
- Wrote hands-on labs in basic circuitry, hardware input, motor control and coached groups through final projects.
- Hired and trained an undergraduate teaching assistant to assist with course activities and grading.
- Granted $3000 from Caltech in funding for larger iteration of the course for the 2024 cycle

Caltech Robotics and Computer Science Pasadena, California
Teaching Assistant JAN 2022 — Present

- Designed a differential drive robot as Robotics Head Teaching Assistant for a 35 student diversity, equity and inclusion program.
- Held office hours and wrote exam materials for classes of 100+ students in Experimental Robotics (ROS2: Python), Kinematics (ROS2: Python), Data Structures (Java), Software Design (C), and Computing Systems (x86 Assembly).

AWARDS

NSF Graduate Research Fellowship APR 2024
Awarded during my senior year at Caltech for my PhD at Georgia Tech. Atlanta, Georgia

IRIM Graduate Research Fellowship APR 2024
Fellowship offered by Georgia Tech's Institute of Robotics and Intelligent Machines. Atlanta, Georgia

Thomas A. Tisch Prize for Undergraduate Teaching in Computing and Mathematical Sciences APR 2024
Awarded for development and instruction of new robotics course at Caltech. Pasadena, California

Kiyo and Eiko Tomiyasu Research Fellowship JUL 2023
Awarded a fully funded summer undergraduate research at the Burdick Group. Pasadena, California

Housner Fund DEC 2023
Granted $3300 to develop a Caltech robotics course, supporting increased student interest. Pasadena, California

SELECTED PROJECTS

- **Human-Robot Manipulation:** built and programmed 5-DOF manipulator robot to play *Carrom*, a game similar to pool, in a dynamic environment with human opponents.
- **ReAct-Based Semantic Wikipedia Scraper**: combined chain-of-thought + action (*ReAct*) with semantic Wikipedia search, improved on previously published baselines
- **AI Diagnostician:** Trained large vision models to predict pathologies from "CheXpert" chest X-rays.
- **Exploration-Based Planner:** Wrote an exploration algorithm and simulative environment to prioritize information gain of an unknown space with probabilistic sensor feedback
- **Robot Planning Computation Cost Analysis:** Built a D* planner and compared computational efficiency with A* and Dijkstra for replanning on large graphs.
- **Quadruped Kinematics:** Wrote forward and inverse kinematics to simulate a walking quadrupedal robot designed in Solidworks using ROS1.
- **Hierarchical Quadruped Planner:** Wrote hierarchical A* planner for quadrupedal locomotion on diverse terrains.
- **Mobile Robotics:** Implemented localization, obstacle avoidance, and odometry algorithms on hardware.

GENERAL SKILLS

- **Programming Languages:** `Python, C/C++`
- **Software:** JaX, Brax, Flax, PyTorch, OpenCV, CVXPY, MuJoCo, ROS1&2, (Ignition) Gazebo, Linux, Git, CMake, MATLAB/Simulink, Mathematica, Solidworks, SimpleFOC
- **Hardware:** Wandercraft Atalante Exoskeleton, Westwood Robotics BRUCE, DEVCOM GVR Bot